INCH-POUND

MIL -PRF-19500/647
18 April 1997

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON,
POWER RECTIFIER, ULTRAFAST,
TYPES 1N6778 THROUGH 1N6779
JANTX, JANTXV and JANS

This specification is approved for use by all Depart
ments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, ultrafast, power rectifier diodes. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-257).

1.3 Maximum ratings.

Types	V_{RWM} I_D= 10 µA dc	I_F 1/ T_C = +100°C	I_{FSM} t_p = 8.3ms	$R_{\theta JC}$	$R_{\theta JA}$	T_{STG} and T_{OP}
	V dc	A dc	A(pk)	°C/W	°C/W	°C
1N6778	400	15	140	1.8	40	
1N6779	600	15	140	1.8	40	-65 to +150

1/ Derate at 300 mA/°C above T_C = +100°C.

1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics are at +25°C.

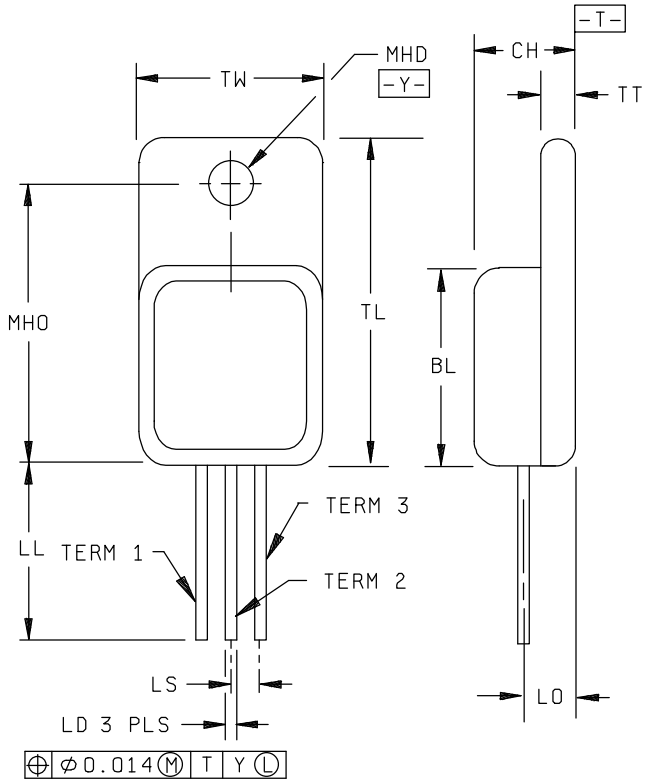
Types	V_{F1} I_F = 15 A dc	V_{F2} I_F = 8 A dc	I_{R1} V_R = 0.8 V_{RWM} (See 1.3)	I_{R2} V_R = 0.8 V_{RWM} (See 1.3) T_C = +100°C	t_{rr}	C_J V_R = 5 V f = 1 MHz
	V dc	V dc	µA dc	µA dc	ns	pF
1N6778					60	
1N6779	1.60	1.40	10	1,000	60	300

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center, Columbus, ATTN: DSCC-VA, 3990 East Broad Street, Columbus, Ohio 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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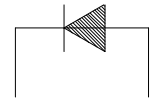


Dimensions				
Symbol	Inches		Millimeters	
	Min	Max	Min	Max
BL	10.4	10.9	0.410	0.430
CH	6.32	6.60	0.249	0.260
LD	0.89	1.14	0.035	0.045
LL	12.70	19.05	0.500	0.750
LO	3.05 typ		0.120 typ	
LS	5.08 bsc		0.200 bsc	
MHD	3.55	3.80	0.140	0.150
MHO	13.4	13.6	0.527	0.537
TL	16.4	16.9	0.645	0.665
TT	1.02	1.27	0.040	0.050
TW	10.4	10.7	0.410	0.420

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only. See 3.3.
3. All terminals are isolated from case.

Configuration

Terminal	Description
1	Cathode
2	Anode
3	No connection



1 2

FIGURE 1. Physical dimensions and configuration TO-257.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent Al_2O_3(ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages. The preferred measurements used herein is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of a conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

3.3.1 Lead material and finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 of table II of MIL-PRF-19500 and 100 percent DC testing in accordance with group A, subgroup 2 herein.

3.3.2 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and table III herein.

4.3 Screening (JANTX, JANTXV, and JANS levels). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3C 1/	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9 and 10	Not applicable	Not applicable
11	I_{R1} and V_{F1}	I_{R1} and V_{F1}
12	See 4.3.1, t = 120 hours	See 4.3.1, t = 48 hours
13	Subgroups 2 and 3 of table I herein; VF1 and IR1; $DI_{R1} \leq 100$ percent of initial value or ±250 nA, whichever is greater; $DV_{F1} \leq \pm 100$ mV.	Subgroup 2 of table I herein; VF1 and IR1; $DI_{R1} \leq 100$ percent of initial value or ±250 nA, whichever is greater; $DV_{F1} \leq \pm 100$ mV.

1/ Thermal impedance shall be performed any time before screen 13.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

MIL-STD-750 method 1038, test condition A. T_C = +125°C; V_R = 0.8 of rated V_{RWM} (see 1.3).

4.3.2 Thermal impedance ($Z_{\theta JX}$) measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) and conditions for $Z_{\theta JX}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition.

4.3.2.1 Thermal impedance ($Z_{\theta JX}$) measurements for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record data $Z_{\theta JX}$). Derived conditions limits and thermal response curve shall be supplied to the qualifying activity on the qualification lot prior to qualification approval. Measurement conditions shall be in accordance with 4.4.1.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. I_M measure current ---------- 15 mA

b. I_H forward heating current ---------- 10-50 A

c. t_H heating time ---------- 200 ms

d. t_{MD} measurement delay time ---------- 35 µs

e. V_H heating voltage ---------- 1 V

The maximum limit for $Z_{\theta JX}$ under these conditions are $Z_{\theta JX}$(max) = 1.6°C/W

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
4	1037	I_F or I_O = 1.25 A to 10 A; ΔT_J = +85°C, minimum for 2,000 cycles minimum.

4.4.2.2 Group B inspection, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
3	1037	I_F or I_O = 1.25 A to 10 A; ΔT_J = +85°C, minimum for 2,000 cycles minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
2	2036	Test condition A, 5 pounds, t = 15 seconds ± 3 seconds.
6	1037	I_F or I_O = 1.25 A to 10 A; ΔT_J = +85°C, minimum for 6,000 cycles minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps and footnotes of table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		1.6	°C/W
Breakdown voltage 1N6778 1N6779	4022	I_R = 10 µA dc, pulsed 1/	V_{BR}	 400 600		V dc
Forward voltage	4011	I_F = 15 A dc, pulsed 1/	V_{F1}		1.60	V dc
		I_F = 8 A dc, pulsed 1/	V_{F2}		1.40	V dc
Reverse leakage current	4016	DC method; pulsed 1/ V_R = 0.8 of V_{RWM} (see 1.3)	I_{R1}		10	µA dc
Subgroup 3						
High temperature operation:		T_C = +100°C				
Reverse leakage current	4016	DC method; pulsed 1/ V_R = 0.8 of V_{RWM} (see 1.3)	I_{R2}		1.0	mA dc
Low-temperature operation:		T_A = -55°C				
Forward voltage	4011	I_F = 15 A dc, pulsed 1/	V_{F3}		1.80	V dc
Subgroup 4						
Scope display evaluation 2/						
Reverse recovery time measurements	4023	Condition B; I_F = 1.0 A, di/dt = 50 A/µs	t_{rr}		60	ns
Subgroups 5 and 6						
Not applicable						
Subgroup 7						
Junction capacitance	4001	V_R = 5 V dc; f = 1.0 MHz	C_J		300	pF

1/ Pulse test: pulse width = 300 µs, duty cycle ≤ 2 percent.

2/ The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 50 to 100 µA/division and 50 to 100 V/division. Reverse current over the knee shall be at least 500 µA. Each device may exhibit a slightly rounded characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.

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TABLE II. Groups A, B, C, and E electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Forward voltage	4011	I_F = 15 A dc pulsed	V_{F1}		1.60	V dc
2	Reverse leakage current	4016	V_R = 0.8 of V_{RWM} (see 1.3) DC method, pulsed	I_{R1}		10	µA dc
3	Forward voltage	4011	I_F = 15 A dc pulsed	DV_{F1}		±100 mV dc from initial reading	
4	Reverse leakage current	4016	V_R = 0.8 of V_{RWM} (see 1.3) DC method, pulsed	ΔI_{R1}		100 percent or ±250 nA dc whichever is greater	
5	Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		1.6	°C/W

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1 and 2.

b. Subgroup 4, see table II herein, steps 1, 2, 3, 4, and 5.

c. Subgroup 5, see table II herein, steps 1, 2, 3, and 4.

2/ The electrical measurements for table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

a. Subgroup 3, see table II herein, steps 1, 2, and 5.

a. Subgroup 6, see table II herein, steps 1 and 2.

3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:

a. Subgroups 2 and 3, see table II herein, steps 1 and 2 for all levels.

a. Subgroup 6, see table II herein, steps 1, 2, and 5 for all levels.

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TABLE III. Group E inspection (all quality levels) for qualification only. 1/

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Thermal shock (temperature cycling)	1051	500 cycles	
Hermetic seal	1071		
Fine leak		Test condition H 5 x 10^{-7} atm cc/s	
Gross leak		Test condition C or K	
Electrical measurements		See table II herein, steps 1 and 2	
Subgroup 2			22 devices c = 0
Steady-state reverse bias	1038	Test condition A t = 1,000 hours, T_C = +125°C V_R = 0.8 of rated V_{RWM} (see 1.3)	
Electrical measurements		See table II herein, steps 1 and 2	
Subgroup 3			3 devices c = 0
Destructive physical analysis	2101		
Subgroup 4			22 devices c = 0
Thermal resistance	3101	see 4.5.2; $R_{\theta JC}$ = 1.8°C/W	
Subgroup 5			15 devices c = 0
Barometric pressure (reduced)	1001	V_R = rated V_{RWM} (see 1.3)	

1/ For initial design and process change verification only (one time testing).

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4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3101 of MIL-STD-750. The maximum limits for $R_{\theta JC}$(max) shall be 2.0°C/W for devices in the 2 pin case style. The following parameter measurements shall apply:

a. I_M measure current ---------- 15 mA

b. I_H forward heating current ---------- 10-50 A

c. t_H heating time ---------- Steady state (see MIL-STD-750, method 3101)

d. t_{MD} measurement delay time ---------- 35 µs

e. V_H heating voltage ---------- 1 V

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging must be in accordance with MIL-PRF-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of documents referenced (see 2.2.1).

c. Lead finish or formation may be specified (see 3.3.1).

d. Type designation and product assurance level.

6.3 Cross reference list. Parts covered by this specification may be used to replace the following commercial Part of Identifying Numbers (PIN):

Preferred types	Commercial types
JANTX1N6778 JANTX1N6779	1N6778 1N6779
JANTXV1N6778 JANTXV1N6779	1N6778 1N6779

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961- 1894)

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3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, ULTRAFAST, TYPES 1N6778 THROUGH 1N6779, JANTX, JANTXV, AND JANS		
4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)		
5. REASON FOR RECOMMENDATION		
6. SUBMITTER		
a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
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